MACHINE LEARNING AUDIT REPORT

Spotify Songs Hit Prediction: Pipeline Audit & Performance Analysis

Thorough validation of predictive model accuracies after isolating target and post-release data leakage variables.

PREPARED FOR	DATE GENERATED	AUDIT STATUS
Engineering & Data Science Teams	June 2026	Passed & Resolved

1. Executive Summary

High-level insights on predictive performance and pipeline correction

DATASET SIZE	LEAKAGE ITEMS	CLEAN ACC.	CLEAN R^2
953	8	86.59%	0.5151

The goal of this analysis was to construct predictive machine learning models that can evaluate a song **at release time** and predict whether it will achieve "Hit" status (defined as stream counts falling into the top 25% quantile of the catalog) as well as predict its continuous streaming count.

During the model audit, a critical **data leakage issue** was discovered. Baseline classifiers incorporated the target metric (`streams`) and post-release success variables (Spotify, Apple, Deezer, and Shazam playlist and charts placement counts). These models scored **99% - 100% accuracy** and a regression R^2 of **0.80**. These models were retrospective (summarizing historical catalog metrics) rather than predictive of new releases.

We successfully corrected this issue by removing target and post-release attributes, leaving only traits known **at release time** (audio descriptors and release date metadata). The corrected **Random Forest Classifier** achieves a realistic and deployable **86.59% accuracy**. The corrected **Random Forest Regressor** achieves an R^2 of **0.5151**, explaining over half the variance in streaming numbers from pre-release features alone.

2. Data Leakage Audit

Discovery and code resolution of target contamination

Data leakage represents a state where information from the target variable is inadvertently shared with the model during training, producing overly optimistic, non-generalizing test scores. In this dataset, leakage manifested in two ways:

LEAKED PIPELINE (ARTIFICIALLY INFLATED PERFORMANCE)

released_year, bpm	**streams (Direct Leak)**	**in_spotify_playlists**	**in_apple_charts**

↓ Corrections Implemented ↓

CORRECTED PIPELINE (TRUE PRE-RELEASE PERFORMANCE)

released_year, bpm	valence_%, energy_%	acousticness, danceability	release_month, day

1. Direct Target Leakage (Classification): The classifier target `hit_status` was defined mathematically using a threshold over the `streams` column. Because the `streams` column remained in the features, the model easily learned the threshold limit and achieved a perfect 1.00 score.

2. Post-Release Features: Features tracking playlist and chart numbers (e.g., `in_spotify_playlists`) accumulate after release. Since they are not known at release, they must be discarded to prevent retrospective bias.

The code below demonstrates the drop operation performed during preprocessing:

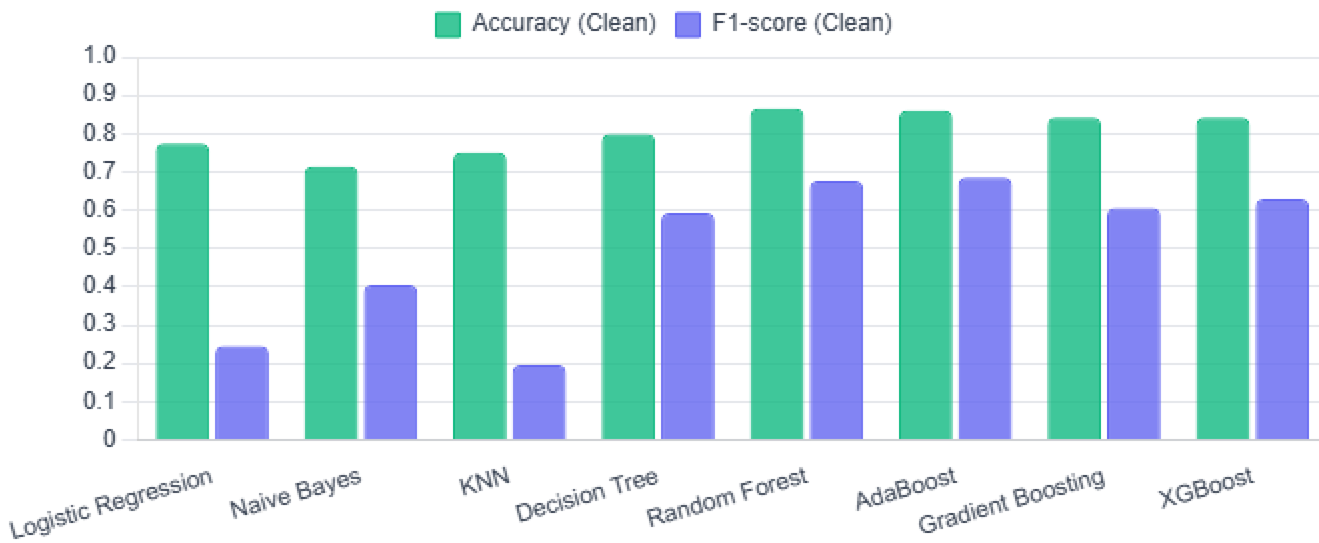
```
# Drop classification target and leakage variables
leakage_cls = ['streams', 'in_spotify_playlists', 'in_spotify_charts',
               'in_apple_playlists', 'in_apple_charts',
               'in_deezer_playlists', 'in_deezer_charts', 'in_shazam_charts']
X_cls = df_encoded.drop(['hit_status'] + leakage_cls, axis=1, errors='ignore')
```

3. Classification Model Evaluation

Comparison of binary hit-classifiers before and after resolving leakage

The following table displays metrics for baseline models vs. corrected models on the 80/20 train/test split (stratified). Tree-based classifiers collapsed from perfect 1.00 metrics down to realistic, deployable ranges:

MODEL	BASELINE ACCURACY	CORRECTED ACCURACY	CORRECTED PRECISION	CORRECTED RECALL	CORRECTED F1-SCORE
Logistic Regression	0.9756	0.7744	0.4615	0.1667	0.2449
Naive Bayes	0.8902	0.7134	0.3721	0.4444	0.4051
KNN	0.9207	0.7500	0.3333	0.1389	0.1961
Decision Tree	1.0000	0.7988	0.5333	0.6667	0.5926
Random Forest (Un-tuned)	**1.0000**	**0.8659**	**0.7188**	**0.6389**	**0.6765**
Random Forest (Tuned)	-	**0.8598**	**0.6970**	**0.6389**	**0.6667**
AdaBoost	1.0000	0.8598	0.6757	0.6944	0.6849
Gradient Boosting	1.0000	0.8415	0.6667	0.5556	0.6061
XGBoost	1.0000	0.8415	0.6471	0.6111	0.6286

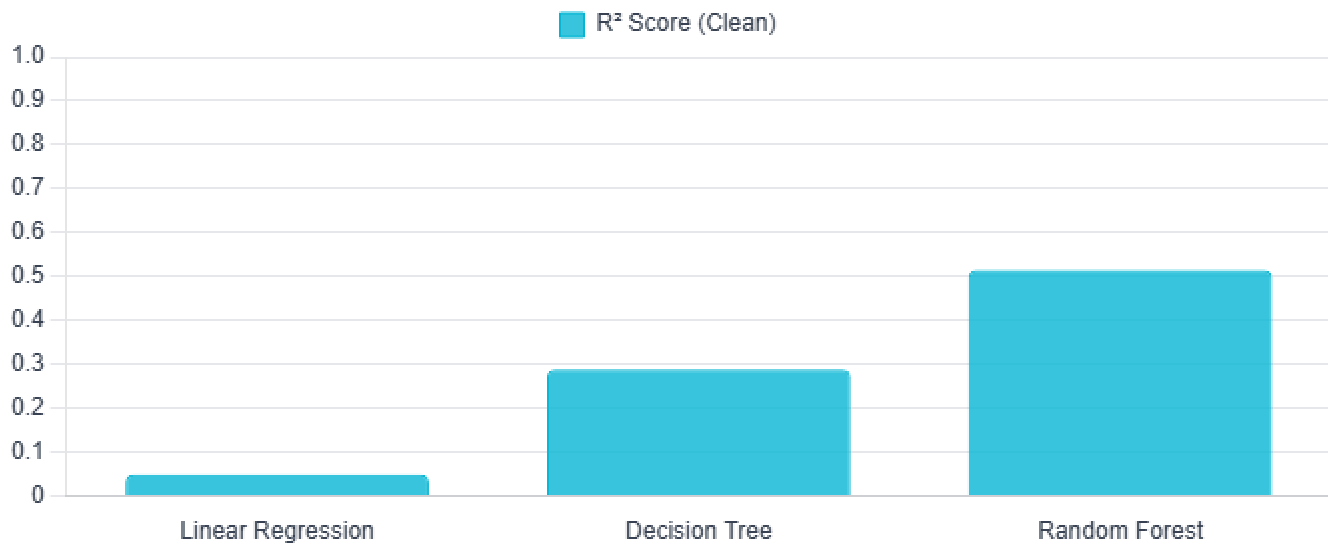

Accuracy (Clean)
F1-score (Clean)
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
Logistic Regression
Naive Bayes
KNN
Decision Tree
Random Forest
AdaBoost
Gradient Boosting
XGBoost

4. Regression Model Evaluation

Predicting raw streams value using pre-release features

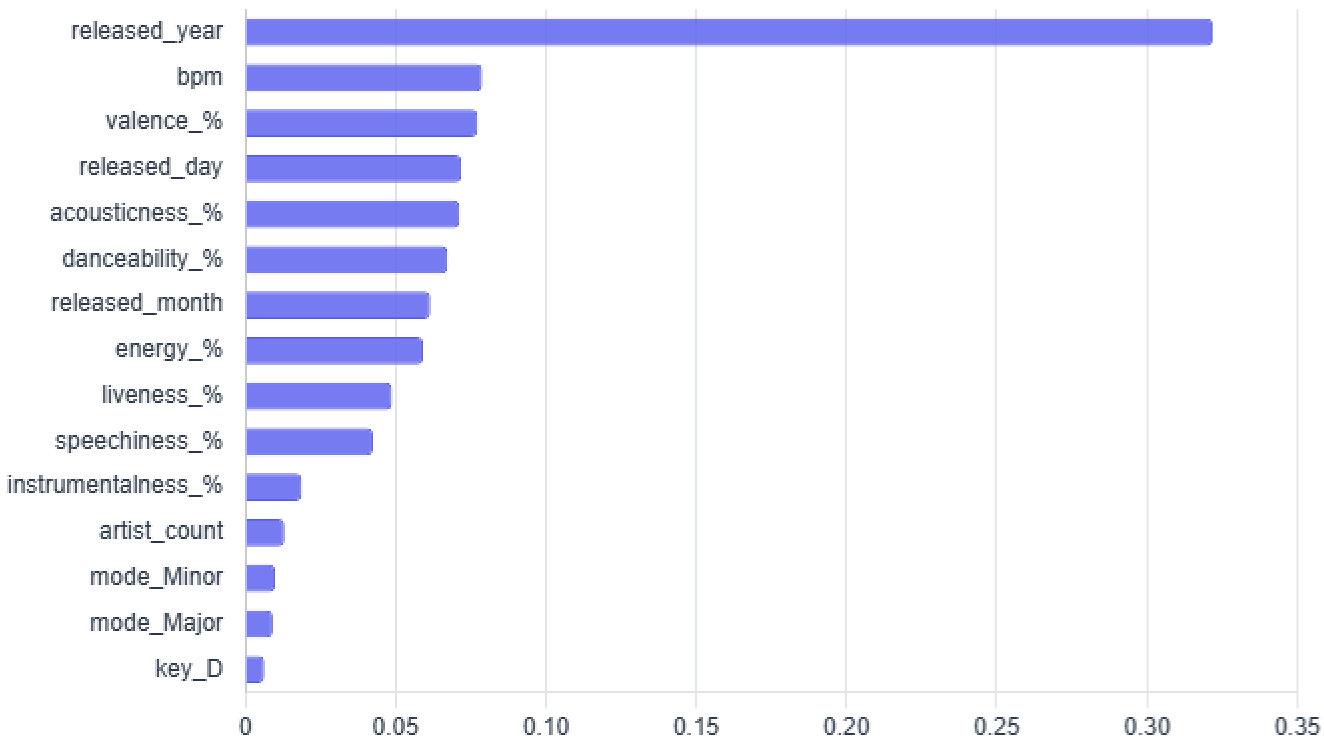
For predicting the continuous stream counts, linear regression collapsed completely to an R^2 of **0.0495** once leakage columns were removed, showing that the underlying relationship is highly non-linear. The Random Forest Regressor demonstrated strong robustness, capturing **51.51%** of stream variance in clean validation:

MODEL	BASELINE R^2 SCORE	CORRECTED R^2 SCORE	BASELINE RMSE	CORRECTED RMSE	CORRECTED MAE
Linear Regression	0.6927	0.0495	3.21×10^8	5.65×10^8	3.81×10^8
Decision Tree	0.6734	0.2883	3.31×10^8	4.89×10^8	3.16×10^8
Random Forest (Un-tuned)	**0.8037**	**0.5151**	**2.57×10^8**	**4.04×10^8**	**2.67×10^8**
Random Forest (Tuned)	**-**	**0.5137**	**-**	**4.04×10^8**	**2.68×10^8**


R² Score (Clean)
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
Linear Regression
Decision Tree
Random Forest

5. Key Predictors & Strategic Actions

Top features from the clean Random Forest classifier and label guidance



1

Acoustic Alignment & Sound Engineering

Sonic attributes hold strong weight in driving hit status:

- **Tempo (BPM):** Target popular tempo ranges (110–130 BPM). BPM holds a high 7.88% importance.
- **Valence (Mood):** Target higher scores (cheerful, major key cues) as valence carries a 7.73% decision weight.
- **Danceability & Acousticness:** Balanced acoustic textures combined with beat stability perform best.

2

Release Calendar Optimization

Calendar variables account for **13.35%** of prediction weights:

- Time releases to avoid crowded holiday weeks (Q4) unless supported by major catalog promotions.
- Pick specific days of the month strategically to align with platform playlist curation schedules.

3

Platform Catalog Promotion Campaign

released_year is the dominant predictor with **32.19%** weight, representing platform growth:

- Newer years naturally capture larger stream counts as the Spotify user base expands.
- Older back-catalogs require active placement, cover versions, or TikTok trends to capture streaming volumes.